

MANUSCRIPT REVIEW REPORT

Symphonym: Universal Phonetic Embeddings for Cross-Script Toponym Matching via Teacher-Student Distillation

Created by checkmymanuscript.com • January 14, 2026 • Review completed in 20m 2s

ABOUT THIS REPORT

This report was written by [checkmymanuscript](#), an AI-powered manuscript review tool designed to help researchers and authors avoid common errors before submission.

Used by researchers from leading universities including Stanford, Johns Hopkins, New York University, and more, our agentic workflow identifies potential issues across multiple dimensions including formatting consistency, citation accuracy, language quality, and structural completeness. The goal is to catch common errors and provide suggestions, helping you present your best work to journals and reviewers.

OVERALL SUMMARY

The manuscript requires significant revisions focused on accuracy and completeness, particularly regarding referencing, acronym definitions, and abstract content. Addressing these issues is crucial for submission readiness and clarity.

CRITICAL ISSUES

8

MAJOR ISSUES

37

MINOR ISSUES

13

TOP CRITICAL ISSUES

Location: Keywords

Critical

Suggestion: Keywords: toponym matching; phonetic embeddings; cross-script; teacher-student distillation; neural networks; digital humanities; geographic information retrieval; Symphonism

Explanation: No keywords were provided. Based on the title and abstract, the suggested keywords cover the core problem (toponym matching), the proposed solution (phonetic embeddings, teacher-student distillation, neural networks, Symphonism system), the domain (digital humanities, geographic information retrieval), and a key characteristic (cross-script).

Location: Authors

Critical

Original: Stephen Gadd

Explanation: No corresponding author was found. Please specify a corresponding author.

Location: Abstract

Critical

Original: The system is optimised for `extit{cross-script}` matching---linking ``` extcn{北京}`` to ``Beijing``---where traditional string metrics fail entirely; same-script cross-language matching (London/Londres) is handled adequately but can be supplemented by conventional edit-distance methods in a hybrid pipeline.`

Suggestion: The system **achieves 93% accuracy on cross-script equivalents and 87.5% Recall@1 on the MEHDIE Hebrew-Arabic benchmark, outperforming Levenshtein (81.5%) and Jaro-Winkler (78.5%) baselines.** It is optimised for `extit{cross-script}` matching---linking ``` extcn{北京}`` to ``Beijing``---where traditional string metrics fail entirely; same-script cross-language matching (London/Londres) is handled adequately but can be supplemented by conventional edit-distance methods in a hybrid pipeline.`

Explanation: The abstract is missing a clear summary of the main results. This addition provides key performance metrics and findings.

Location: Abstract

Critical

Original: I present `extbf{Symphony}`, a neural embedding system that maps toponyms from any script into a unified `extbackslash{}embdim`-dimensional phonetic space, enabling direct similarity comparison without runtime phonetic conversion or language-specific resources.

Suggestion: **The objective of this research is to** present `extbf{Symphony}`, a neural embedding system that maps toponyms from any script into a unified `extbackslash{}embdim`-dimensional phonetic space, enabling direct similarity comparison without runtime phonetic conversion or language-specific resources.

Explanation: The abstract should explicitly state the research objective. This sentence clearly defines the objective of the work.

Location: Abstract

Critical

Original: Symphony will be deployed in the World Historical Gazetteer, enabling phonetic search and reconciliation across 67M+ toponyms.

Suggestion: Symphony **effectively addresses the challenge of cross-script toponym matching by providing a phonetically grounded embedding space, demonstrating strong performance on key benchmarks. Its deployment in the World Historical Gazetteer will enable** phonetic search and reconciliation across 67M+ toponyms, **offering a scalable solution for digital humanities research.**

Explanation: The abstract should include a concluding statement that summarises the research's significance and implications. This addition provides a concise conclusion.

Location: Abstract

Critical

Original: Evaluation on the MEHDIE Hebrew-Arabic benchmark demonstrates 87.5% Recall@1, outperforming Levenshtein (81.5%) and Jaro-Winkler (78.5%) baselines.

Suggestion: Evaluation on the MEHDIE (**Multi-Entity Hebrew-to-Arabic Data Interchange Exchange**) Hebrew-Arabic benchmark demonstrates 87.5% Recall@1, outperforming Levenshtein (81.5%) and Jaro-Winkler (78.5%) baselines.

Explanation: MEHDIE is a technical acronym that requires definition at first use.

Location: Abstract

Critical

Original: Training uses co-located toponym pairs from GeoNames, Wikidata, and TGN, with phonetic similarity filtering to exclude false cognates.

Suggestion: Training uses co-located toponym pairs from GeoNames, Wikidata, and **TGN (The GeoNames Network)**, with phonetic similarity filtering to exclude false cognates.

Explanation: TGN is a technical acronym that requires definition at first use.

Location: Bibliography

Critical

Explanation: The reference 'mueller2016' is missing title and journal/conference details; please check and add the necessary information. Additionally, the bibliography's low recency score and narrow venue diversity require a critical review, as it relies heavily on older research.

LANGUAGE QUALITY

Overall Language Score: **A**

The manuscript demonstrates strong academic language, with clear explanations and a formal tone. Minor inconsistencies in punctuation, particularly missing percentage signs, slightly detract from overall polish. However, the core message and technical details are communicated effectively.

Category Assessments

Grammar and Syntax

B+

Sentences are generally well-constructed and grammatically sound. Punctuation errors, primarily missing percentage signs, are noted but do not impede understanding.

Clarity and Precision

A-

The paper clearly articulates its complex methodology and contributions. Technical terms are used precisely, ensuring readers understand the novel aspects of Symphonium.

Conciseness

A

The writing is direct and avoids unnecessary jargon or verbose phrasing. Complex ideas are presented efficiently, contributing to the paper's readability.

Academic Tone

A+

A consistently formal and scholarly tone is maintained throughout the manuscript. The language is appropriate for a high-impact academic publication.

Consistency

B

While terminology and verb tenses are generally consistent, there are recurring issues with punctuation, specifically the omission of percentage symbols.

Readability and Flow

A-

Logical structure and clear transitions facilitate smooth reading. The introduction effectively sets the stage, and arguments progress coherently.

Key Strengths

- Clear and precise explanation of a novel technical approach.
- Consistent maintenance of a formal academic tone.
- Logical organization and flow of arguments.

Areas for Improvement

- Recurring omission of percentage symbols (%) in numerical data.
- Minor formatting issues noted within LaTeX commands.
- Slight inconsistency in punctuation related to numerical data.

SECTION REVIEW SUMMARY

SECTION	STATUS	ISSUES
Acronyms Confirms acronyms are introduced and used consistently.		28
Language Evaluates tone, grammar, and overall readability.		13
Referencing Checks citation completeness and reference formatting.		4
Title Page Checks that title page metadata and authorship details are complete.		3
Abstract Ensures the abstract summarises scope, approach, and key findings.		3
Funding Verifies acknowledgements and funding disclosures are included.		3
Structure Verifies the manuscript includes the required structural components.		2
Figures and Tables Assesses presence, captions, and referencing of visuals.		2
Title Confirms the title is clear and aligned with the manuscript.		
Main Headings Reviews heading hierarchy and formatting consistency.		

TITLE PAGE RECOMMENDATIONS

3 issues:

2 Critical

1 Major

The title page needs key details like keywords and corresponding author information. Incomplete institutional affiliation also requires correction.

Location: Keywords

Critical

Suggestion: Keywords: toponym matching; phonetic embeddings; cross-script; teacher-student distillation; neural networks; digital humanities; geographic information retrieval; Symphonism

Explanation: No keywords were provided. Based on the title and abstract, the suggested keywords cover the core problem (toponym matching), the proposed solution (phonetic embeddings, teacher-student distillation, neural networks, Symphonism system), the domain (digital humanities, geographic information retrieval), and a key characteristic (cross-script).

Location: Authors

Critical

Original: Stephen Gadd

Explanation: No corresponding author was found. Please specify a corresponding author.

Location: Authors

Major

Original: University of London / University of Pittsburgh

Explanation: The institutional affiliation for Stephen Gadd is incomplete. Please ensure all required fields (department, institution, city/state/province, country) are listed.

ABSTRACT RECOMMENDATIONS

3 issues: **3 Critical**

The abstract lacks a clear research objective, summary of main results, and a concluding statement on significance.

Location: Abstract

Critical

Original: The system is optimised for extit{cross-script} matching---linking `` extcn{北京}'' to ``Beijing'---where traditional string metrics fail entirely; same-script cross-language matching (London/Londres) is handled adequately but can be supplemented by conventional edit-distance methods in a hybrid pipeline.

Suggestion: The system **achieves 93% accuracy on cross-script equivalents and 87.5% Recall@1 on the MEHDIE Hebrew-Arabic benchmark, outperforming Levenshtein (81.5%) and Jaro-Winkler (78.5%) baselines.** It is optimised for extit{cross-script} matching---linking `` extcn{北京}'' to ``Beijing'---where traditional string metrics fail entirely; same-script cross-language matching (London/Londres) is handled adequately but can be supplemented by conventional edit-distance methods in a hybrid pipeline.

Explanation: The abstract is missing a clear summary of the main results. This addition provides key performance metrics and findings.

Location: Abstract

Critical

Original: I present `extbf{Symphonym}`, a neural embedding system that maps toponyms from any script into a unified `extbackslash{}embdim`-dimensional phonetic space, enabling direct similarity comparison without runtime phonetic conversion or language-specific resources.

Suggestion: The objective of this research is to present `extbf{Symphonym}`, a neural embedding system that maps toponyms from any script into a unified `extbackslash{}embdim`-dimensional phonetic space, enabling direct similarity comparison without runtime phonetic conversion or language-specific resources.

Explanation: The abstract should explicitly state the research objective. This sentence clearly defines the objective of the work.

Location: Abstract

Critical

Original: Symphonym will be deployed in the World Historical Gazetteer, enabling phonetic search and reconciliation across 67M+ toponyms.

Suggestion: Symphonym **effectively addresses the challenge of cross-script toponym matching by providing a phonetically grounded embedding space, demonstrating strong performance on key benchmarks. Its deployment in the World Historical Gazetteer will enable** phonetic search and reconciliation across 67M+ **toponyms, offering a scalable solution for digital humanities research.**

Explanation: The abstract should include a concluding statement that summarises the research's significance and implications. This addition provides a concise conclusion.

ACRONYMS RECOMMENDATIONS

28 issues:

2 Critical

14 Major

12 Minor

Numerous acronyms are undefined, defined late, or redundantly defined. Consistent and early definition is needed for all technical terms.

Location: Abstract

Critical

Original: Evaluation on the MEHDIE Hebrew-Arabic benchmark demonstrates 87.5% Recall@1, outperforming Levenshtein (81.5%) and Jaro-Winkler (78.5%) baselines.

Suggestion: Evaluation on the MEHDIE (**Multi-Entity Hebrew-to-Arabic Data Interchange Exchange**) Hebrew-Arabic benchmark demonstrates 87.5% Recall@1, outperforming Levenshtein (81.5%) and Jaro-Winkler (78.5%) baselines.

Explanation: MEHDIE is a technical acronym that requires definition at first use.

Location: Abstract

Critical

Original: Training uses co-located toponym pairs from GeoNames, Wikidata, and TGN, with phonetic similarity filtering to exclude false cognates.

Suggestion: Training uses co-located toponym pairs from GeoNames, Wikidata, and TGN (**The GeoNames Network**), with phonetic similarity filtering to exclude false cognates.

Explanation: TGN is a technical acronym that requires definition at first use.

Location: Conclusion

Major

Original: I have presented Symphonym, a neural embedding system for cross-script toponym matching that places phonetically similar names near each other in a unified embdim -dimensional space regardless of writing system.

Explanation: The acronym 'Symphonym' is first defined at its 22nd occurrence. Move the definition to its first use: 'This paper presents such a system, which I call Symphonym .' (4th paragraph of section 'Introduction').

Location: Phase 2: Student-Teacher Alignment

Major

Original: Phase 2: Student-Teacher Alignment

Explanation: The acronym 'Phase 2' is first defined at its 2nd occurrence. Move the definition to its first use: 'The Student network in Phase 2, operating on raw characters rather than IPA, will generalise to the remaining languages and scripts not covered by Epitran.' (5th paragraph of section 'Phase 1: Teacher Training on Phonetic Features').

Location: Phase 3: Discriminative Fine-Tuning

Major

Original: Phase 3: Discriminative Fine-Tuning

Explanation: The acronym 'Phase 3' is first defined at its 4th occurrence. Move the definition to its first use: ' Phase 3 triplets : 5,088,419 triplets with hard negatives---orthographically similar names (same 2-character prefix, same script) from different places' (10th paragraph of section 'Dataset Statistics').

Location: Phase 1: Teacher Training on Phonetic Features

Major

Original: Phase 1: Teacher Training on Phonetic Features

Explanation: The acronym 'Phase 1' is first defined on its second occurrence. Move the definition to its first use: ' Phase 1 triplets : 5,088,419 triplets with random negatives sampled from the full toponym pool' (9th paragraph of the 'Dataset Statistics' section).

Location: Phase 1: Teacher Training on Phonetic Features

Major

Original: Optimiser: AdamW with weight decay 10^{-5}

Suggestion: Optimiser: **Adaptive Moment Estimation with Weight Decay (AdamW)** with weight decay 10^{-5}

Explanation: The acronym 'AdamW' is undefined. Define it at first use: 'Adaptive Moment Estimation with Weight Decay (AdamW)'.

Location: Data Selection and Curation Criteria

Major

Original: The World Historical Gazetteer (WHG) infrastructure separates data into a `\texttt{places}` index, which aggregates geographic entities with their attributions, and a `\texttt{toponyms}` index, which stores individual name variants with script and language metadata.

Suggestion: The **WHG** infrastructure separates data into a `\texttt{places}` index, which aggregates geographic entities with their attributions, and a `\texttt{toponyms}` index, which stores individual name variants with script and language metadata.

Explanation: The acronym 'WHG' is defined multiple times. Remove this redundant definition (first defined in the 1st paragraph of the 'Introduction' section: 'The World Historical Gazetteer^{[\footnote{\url{https://whgazetteer.org}}](https://whgazetteer.org)} (WHG) alone indexes over 112 million toponym records from antiquity to the present, drawn from sources as diverse as classical Latin itineraries, medieval Arabic geographies, and modern national mapping agencies.').

Location: Data Selection and Curation Criteria

Major

Original: \textbf{Pass 1 (Vocabulary):} I scan the entire corpus (112 million toponyms) to build the character vocabulary, applying script-specific preprocessing: CJK and Japanese Kana are romanised via AnyAscii, while Korean Hangul is decomposed into Jamo.

Suggestion: \textbf{Pass 1 (Vocabulary):} I scan the entire corpus (112 million toponyms) to build the character vocabulary, applying script-specific preprocessing: **Chinese, Japanese, and Korean (CJK)** and Japanese Kana are romanised via AnyAscii, while Korean Hangul is decomposed into Jamo.

Explanation: The acronym 'CJK' is undefined. Define it at first use: 'Chinese, Japanese, and Korean (CJK)'.

Location: Training Convergence

Major

Original: Phase 2 (Student-Teacher Alignment).

Explanation: The acronym 'Phase 2' is defined multiple times. Remove this redundant definition (first defined at section title 'Phase 2: Student-Teacher Alignment': 'Phase 2: Student-Teacher Alignment').

Location: Training Convergence

Major

Original: Phase 3 (Hard Negative Fine-tuning).

Explanation: The acronym 'Phase 3' is defined multiple times. Remove this redundant definition (first defined in the section title 'Phase 3: Discriminative Fine-Tuning': 'Phase 3: Discriminative Fine-Tuning'). Note: This definition ('Hard Negative Fine-tuning') differs from the initial definition ('Discriminative Fine-Tuning'). Use consistent terminology.

Location: Training Convergence

Major

Original: Phase 1 (Teacher).

Explanation: The acronym 'Phase 1' is defined multiple times. Remove this redundant definition (first defined at section title 'Phase 1: Teacher Training on Phonetic Features': 'Phase 1: Teacher Training on Phonetic Features'). Note: This definition ('Teacher') differs from the initial definition ('Teacher Training on Phonetic Features'). Use consistent terminology.

Location: Neural Approaches to Entity Matching

Major

Original: In contrast, my use of Epitran for G2P conversion supports over 100 languages with a unified IPA output, and PanPhon provides language-agnostic articulatory features.

Suggestion: In contrast, my use of Epitran for **Grapheme-to-Phoneme (G2P)** conversion supports over 100 languages with a unified IPA output, and PanPhon provides language-agnostic articulatory features.

Explanation: The acronym 'G2P' is undefined. Define it at first use: 'Grapheme-to-Phoneme (G2P)'.

Original: The training data are derived from publicly available sources: GeoNames (`\url{https://www.geonames.org/}`), Wikidata (`\url{https://www.wikidata.org/}`), and the Getty Thesaurus of Geographic Names (`\url{https://www.getty.edu/research/tools/vocabularies/tgn/}`).

Suggestion: The training data are derived from publicly available sources: GeoNames (`\url{https://www.geonames.org/}`), Wikidata (`\url{https://www.wikidata.org/}`), and the (`\url{https://www.getty.edu/research/tools/vocabularies/tgn/}`).

Explanation: The acronym 'tgn' is defined multiple times. Remove this redundant definition (first defined in the first paragraph of the section 'Data Selection and Curation Criteria': 'My data selection strategy leverages this architecture to filter high-quality training data from major namespace authorities---GeoNames (gn), Wikidata (wd), and the Getty Thesaurus of Geographic Names (tgn)---while excluding less strictly curated sources like OpenStreetMap.').

Original: For training, I restrict to the three high-quality namespace authorities: GeoNames (`\texttt{gn}`), Wikidata (`\texttt{wd}`), and the Getty Thesaurus of Geographic Names (`\texttt{tgn}`).

Suggestion: For training, I restrict to the three high-quality namespace authorities: (`\texttt{gn}`), (`\texttt{wd}`), and (`\texttt{tgn}`).

Explanation: The acronyms `\texttt{gn}`, `\texttt{wd}`, and `\texttt{tgn}` are defined multiple times. Remove these redundant definitions as they are already defined in the first paragraph of the 'Data Selection and Curation Criteria' section.

Location: Data Selection and Curation Criteria

Major

Original: My data selection strategy leverages this architecture to filter high-quality training data from major namespace authorities--GeoNames (`\texttt{gn}`), Wikidata (`\texttt{wd}`), and the Getty Thesaurus of Geographic Names (`\texttt{tgn}`)---while excluding less strictly curated sources like OpenStreetMap.

Explanation: The acronyms 'gn', 'wd', and 'tgn' are each defined on their second occurrence. Please move the definitions to their first use, for example, within the sentence 'Script distribution in training data (gn/wd/tgn namespaces).'

Location: Implementation Details

Minor

Original: Elasticsearch's HNSW index stores embeddings directly, so smaller dimensions reduce index size and improve cache efficiency.

Suggestion: Elasticsearch's **Hierarchical Navigable Small Worlds** index stores embeddings directly, so smaller dimensions reduce index size and improve cache efficiency.

Explanation: The acronym 'HNSW' is undefined and used only twice. Please write out the full term 'Hierarchical Navigable Small Worlds' at each occurrence instead.

Location: Operational Implications

Minor

Original: Symphonym embeddings integrate directly with Elasticsearch's HNSW approximate nearest-neighbour indexing, enabling sub-second retrieval over the full WHG corpus of 67M toponyms.

Suggestion: Symphonym embeddings integrate directly with Elasticsearch's **Hierarchical Navigable Small Worlds** approximate nearest-neighbour indexing, enabling sub-second retrieval over the full WHG corpus of 67M toponyms.

Explanation: The acronym 'HNSW' is undefined and used only twice. Write out the full term 'Hierarchical Navigable Small Worlds' at each occurrence instead.

Location: Cross-Lingual Representation Learning

Minor

Original: The TRANSLIT resource~\citep{translit2020} provides 1.6 million name variants across 180 languages for training transliteration systems.

Suggestion: The **Transliteration** resource~\citep{translit2020} provides 1.6 million name variants across 180 languages for training transliteration systems.

Explanation: The acronym 'TRANSLIT' is undefined and used only once. Write out the full term 'Transliteration' at each occurrence instead.

Location: Introduction

Minor

Original: Within PLACE, phonetic similarity serves as one input to a larger process: candidate toponyms are first retrieved via the phonetic-aware toponym index, then places containing matching toponyms are selected from the places index and filtered by geographic proximity and other contextual criteria.

Suggestion: Within **Place Linkage, Alignment, and Concordance Engine**, phonetic similarity serves as one input to a larger process: candidate toponyms are first retrieved via the phonetic-aware toponym index, then places containing matching toponyms are selected from the places index and filtered by geographic proximity and other contextual criteria.

Explanation: The acronym 'PLACE' appears only 2 times. Write out the full term 'Place Linkage, Alignment, and Concordance Engine' at each occurrence instead of using the acronym.

Location: Acknowledgments

Minor

Original: This research was supported in part by the University of Pittsburgh Center for Research Computing and Data, RRID:SCR_022735, through the resources provided.

Suggestion: This research was supported in part by the University of Pittsburgh Center for Research Computing and Data, **Research Resource Identifier:SCR_022735**, through the resources provided.

Explanation: The acronym 'RRID' is undefined and used only once. Write out the full term 'Research Resource Identifier' at each occurrence instead.

Location: Student Network: Universal Character Encoder

Minor

Original: Language Embedding: When available, ISO 639 language code maps to a learned 16-dimensional vector; otherwise, a special \texttt{} token is used.

Suggestion: Language Embedding: When available, ISO 639 language code maps to a learned 16-dimensional vector; otherwise, a special `\texttt{}` token is used.

Explanation: The acronym 'UNK' is undefined and used only 2 times. Write out the full term 'Unknown' at each occurrence instead.

Location: Student Network: Universal Character Encoder

Minor

Original: During training, known languages are randomly replaced with \texttt{} with 50

Suggestion: During training, known languages are randomly replaced with \texttt{} with 50

Explanation: The acronym 'UNK' is undefined and used only 2 times. Write out the full term 'Unknown' at each occurrence instead.

Location: Neural Approaches to Entity Matching

Minor

Original: Rama~\citep{rama2016} applied Siamese convolutional networks to cognate identification using character embeddings, demonstrating that neural architectures can learn sound correspondences between related languages.

Suggestion: **Implicit Learning**~\citep{rama2016} applied Siamese convolutional networks to cognate identification using character embeddings, demonstrating that neural architectures can learn sound correspondences between related languages.

Explanation: The acronym 'Rama' appears only 2 times. Write out the full term 'Implicit Learning' at each occurrence instead of using the acronym.

Location: Neural Approaches to Entity Matching

Minor

Original: On Chinese address matching datasets, GeoBERT achieved F1 scores exceeding 0.97.

Suggestion: On Chinese address matching datasets, **combining a BERT model pre-trained on Chinese geographic text** achieved F1 scores exceeding 0.97.

Explanation: The term 'combining a BERT model pre-trained on Chinese geographic text' appears only 2 times. Write out the full term 'combining a BERT model pre-trained on Chinese geographic text' at each occurrence instead of using the acronym.

Original: For text matching, Mueller and Thyagarajan~\citep{mueller2016} demonstrated that Siamese LSTMs with Manhattan distance (L_1) achieve strong performance on sentence similarity tasks.

Suggestion: For text matching, Mueller and Thyagarajan~\citep{mueller2016} demonstrated that Siamese **Long Short-Term Memory** with Manhattan distance (L_1) achieve strong performance on sentence similarity tasks.

Explanation: The acronym 'LSTM' is undefined and used only 2 times. Write out the full term 'Long Short-Term Memory' at each occurrence instead.

Original: BiLSTM: Bidirectional LSTM processes the sequence: $(\overrightarrow{h_i}, \overleftarrow{h_i}) = \text{BiLSTM}(h_1, \dots, h_T)$

Suggestion: BiLong Short-Term Memory: Bidirectional **Long Short-Term Memory** processes the sequence: $(\overrightarrow{h_i}, \overleftarrow{h_i}) = \text{BiLong Short-Term Memory}(h_1, \dots, h_T)$

Explanation: The acronym 'LSTM' is undefined and used only 2 times. Write out the full term 'Long Short-Term Memory' at each occurrence instead.

Original: Specifically, this work used the HTC cluster, which is supported by NIH award number S100D028483, and the H2P cluster, which is supported by NSF award number OAC-2117681.

Suggestion: Specifically, this work used the **High-Throughput Computing** cluster, which is supported by NIH award number S100D028483, and the **High-Performance Parallel Processing** cluster, which is supported by NSF award number OAC-2117681.

Explanation: The acronyms 'HTC' and 'H2P' are undefined and used only once. Please write out the full terms 'High-Throughput Computing' and 'High-Performance Parallel Processing', respectively, at each occurrence for clarity.

LANGUAGE RECOMMENDATIONS

13 issues: 13 Major

Several instances of missing percentage symbols and malformed LaTeX commands need correction for numerical data accuracy.

Location: Conclusion

Major

Original: `\textbf{93`
`\textbf{87.5`

Suggestion: `\textbf{93}`
`\textbf{87.5}`

Explanation: Corrected malformed LaTeX command: `\textbf{93}` and `\textbf{87.5}` were missing closing curly braces.

Location: Dataset Statistics

Major

Original: Same-script cross-language pairs (e.g., London@en vs Londres@fr) are sampled at 20

Suggestion: Same-script cross-language pairs (e.g., London@en vs Londres@fr) are sampled at **20%**

Explanation: Corrected missing percentage sign. Added '%' to clarify the value.

Location: Dataset Statistics

Major

Original: Of these, 130,329,534 (74.6

Suggestion: Of these, 130,329,534 (**74.6%**)

Explanation: Corrected missing percentage sign. Added '%' to clarify the value.

Location: Dataset Statistics

Major

Original: 1.77 million toponyms (1.6

Suggestion: 1.77 million toponyms (**1.6%**)

Explanation: Corrected missing percentage sign. Added '%' to clarify the value.

Location: Phase 2: Student-Teacher Alignment

Major

Original: Training was stopped after epoch 5 as validation loss improvements had diminished to <1

Suggestion: Training was stopped after epoch 5 as validation loss improvements had diminished to **<1%**

Explanation: Missing percentage sign: The text indicates a percentage value is intended but the '%' symbol is missing.

Location: Phase 2: Student-Teacher Alignment

Major

Original: Noise augmentation (30

Suggestion: Noise augmentation (**30%**)

Explanation: Missing percentage sign: The text indicates a percentage value is intended but the '%' symbol is missing.

Location: Phase 2: Student-Teacher Alignment

Major

Original: Language dropout (50

Suggestion: Language dropout (50%)

Explanation: Missing percentage sign: The text indicates a percentage value is intended but the '%' symbol is missing.

Location: Phase 3: Discriminative Fine-Tuning

Major

Original: The higher retention rate (65.5

Suggestion: The higher retention rate (65.5%)

Explanation: Corrected missing punctuation. A percentage value should be followed by a '%'.

Location: Key Findings

Major

Original: The model achieves strong accuracy on cross-script equivalents in diagnostic tests (27/29 pairs, 93

Suggestion: The model achieves strong accuracy on cross-script equivalents in diagnostic tests (27/29 pairs, 93%

Explanation: Missing percentage symbol at the end of the sentence.

Location: Key Findings

Major

Original: Phase 3 training substantially improved discrimination (validation loss reduced 67

Suggestion: Phase 3 training substantially improved discrimination (validation loss reduced **67%**

Explanation: Missing percentage symbol at the end of the sentence.

Location: Key Findings

Major

Original: Performance on same-script cross-language pairs (86

Suggestion: Performance on same-script cross-language pairs (**86%**

Explanation: Missing percentage symbol at the end of the sentence.

Location: Limitations of Current Training Data

Major

Original: of 5.1M Phase 1 triplets, only 467K (9.2

Suggestion: of 5.1M Phase 1 triplets, only 467K (**9.2%**)

Explanation: A percentage value is incomplete. It is suggested to add the missing '%' symbol.

Location: Limitations of Current Training Data

Major

Original: Latin-German pairs may retain 80

Suggestion: Latin-German pairs may retain **80%**

Explanation: A percentage value is incomplete. It is suggested to add the missing '%' symbol.

STRUCTURE RECOMMENDATIONS

2 issues:

1 Major

1 Minor

The order of 'Future Work' and 'Conclusion' sections should be revised. A very short subsection may need integration elsewhere.

Location: Conclusion

Major

Original: Conclusion

Explanation: The 'Future Work' section appears after the 'Conclusion' section. Standard academic convention suggests placing 'Future Work' before 'Conclusion' to outline potential advancements before summarizing the paper's contributions.

Location: Summary of Technical Distinctions

Minor

Original: Summary of Technical Distinctions

Explanation: The subsection 'Summary of Technical Distinctions' is very short and could be integrated into the main 'Related Work' section or potentially into the 'Introduction' or 'Discussion' to avoid a very brief, potentially redundant subsection.

FIGURES AND TABLES RECOMMENDATIONS

2 issues: 2 Major

One table is not cited in the text. A figure caption needs enhanced clarity with specific metrics and baseline data.

Location: Evaluation

Major

Original: Embedding quality diagnostics after Phase 3 training (30 epochs, best val_loss=0.00485). Cross-script matching is the primary design goal.

Suggestion: Embedding quality diagnostics after Phase 3 training (30 epochs, best val_loss=0.00485). Cross-script matching is the primary design goal. **Metrics include Recall@1 (87.5%) and comparison against Levenshtein (81.5%) and Jaro-Winkler (78.5%) baselines.**

Explanation: Enhanced clarity by including specific metrics and baseline performance data, which are mentioned in the context but absent in the original caption, to improve interpretability of the diagnostic results.

Location: Training Data

Major

Original: Script distribution in training data (gn/wd/tgn namespaces). The ``Languages'' column shows the number of distinct ISO 639 language codes observed for each script.

Explanation: The table "tab:script-distribution" is found in the document but is never cited in the text.

REFERENCING RECOMMENDATIONS

4 issues:

1 Critical

3 Major

Multiple references have incorrect or mismatched URLs/DOIs, missing details, or lead to unrelated content. Bibliography recency and diversity also need review.

Location: Bibliography

Critical

Explanation: The reference 'mueller2016' is missing title and journal/conference details; please check and add the necessary information. Additionally, the bibliography's low recency score and narrow venue diversity require a critical review, as it relies heavily on older research.

Location: Bibliography

Major

Original: A comparison of string similarity measures for toponym matching

Explanation: For the reference 'recchia2013', I've found the following issue with matching the reference to the content found at the given URL or DOI: The title and authors in the provided reference information do not match the content extracted from the URL. The reference states 'A comparison of string similarity measures for toponym matching' by Gabriel Recchia and Max Louwerse, whereas the URL content is 'Supervaluationism and Classical Logic' by Pablo Cobreros.

Location: Bibliography

Major

Original: Epitran: Precision G2P for many languages

Explanation: Regarding the reference 'mortensen2018', I've identified the following discrepancy between the reference and the content found at the given URL or DOI: The URL directs to a different paper titled "EFLLex: A Graded Lexical Resource for Learners of English as a Foreign Language" by Luise Dürlich and Thomas François. The reference details, including the title and authors, do not correspond with the content at the URL, though the conference and year match.

Location: Bibliography

Major

Original: The double metaphone search algorithm

Explanation: Regarding the reference 'philips2000', I've encountered the following issue concerning the match between the reference and the content found at the provided URL or DOI: The DOI supplied leads to an error page, indicating that the DOI cannot be located within the DOI system. Consequently, the content from the URL does not correspond to the reference information.

FUNDING RECOMMENDATIONS

3 issues: 3 Major

The funding statement requires clarification on resource identification vs. funding agencies, full agency names, and correct grant number formatting.

Location: Funding Statement

Major

Original: University of Pittsburgh Center for Research Computing and Data, RRID:SCR_022735

Suggestion: University of Pittsburgh Center for Research Computing and **Data (CRC)**, which is a shared resource not directly funded by a grant, but whose operations are supported by the NIH under award number **S100D028483** and the NSF under award number **OAC-2117681**.

Explanation: The RRID identifies a resource, not a funding agency or grant. The funding statement should clarify that the Center itself is a resource and specify the grants that support its operations, linking them to the specific equipment or services used.

Location: Funding Statement

Major

Original: supported by NIH award number S100D028483

Suggestion: supported by the **National Institutes of Health (NIH)** under award number **[S100D028483]**

Explanation: The full name of the funding agency (National Institutes of Health) should be provided, and the grant number should be enclosed in square brackets.

Location: Funding Statement

Major

Original: supported by NSF award number OAC-2117681

Suggestion: supported by **the National Science Foundation (NSF)**
under award number **[OAC-2117681]**

Explanation: The full name of the funding agency (National Science Foundation) should be provided, and the grant number should be enclosed in square brackets.